

Final Year Project

Mumtahir

FYP-I Mid Report

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Chapter 1

Introduction

Mumtahir focuses on revolutionizing how the current system of assignment demos are conducted by integrating a lot of automation and AI into the system. It is aimed to decrease bias allegations, manual labor and reducing conflicts for students and teacher assistants for PF students submitting their assignments in C++.

1.1 Problem Statement

We are developing this system as a tool to alleviate students and assignments checkers from the burden of doing tasks manually. Allocation of time slots and the availability of both parties at the slot is a huge problem. The time suitable to students often is not suitable to the teacher assistant. Students sometimes have to leave the class in the middle or miss classes to wait in queue for the demos. Sometimes the evaluator is busy in the slot agreed upon or has some urgent work which needs to be dealt with and has to reschedule which often puts off the evaluation for a week or more leading to confusion and frustration. All these issues arise just because of time slots and then there are the issues where the evaluator sometimes knows the person they are evaluating and then they get accused of showing positive or negative bias towards the students. This is a serious ethical issue with no clear solution. Then there is the issue of evaluator assigning and uploading marks, analyzing the solution which are a feat on their own and need to be done.

1.2 Motivation

Being students ourselves who had to give countless demos for which we often missed classes got reprimanded for being late, leaving early and gone for most of the class and seeing how inefficient the system is we had the idea to automate and solve as much of these problems as possible. This would take a lot of pressure off of students, evaluators and faculty.

1.3 Problem Solution

We are aiming to create a web app where we will be working on creating a portal which will have student credentials, their info, their facial features. The student will login to the portal and will be checked if there is a face in the frame. Next step would be to verify the person in the frame is the actual student supposed to be evaluated. Next step would be opening the assignment in front of the student and asking them questions about the

assignment or topics related to the assignment which will be chosen by the teacher The answer will be turned into text and checked for accuracy of the answer and student will be graded on the basis of the answer matching.

During the live demo the students facial expressions will be constantly monitored to make sure they are not cheating and are confident in their answers. A report will be generated with transcription of the conversation between student and AI Evaluator with the marks of the student for the demo, the code submitted and the facial expressions throughout the video. The plagiarism for the assignment will be check beforehand and students and teachers will be informed of the cases involved in the act. Submission checking will also be done before the evaluation to make sure the code submitted for the problem corresponds with the problem assigned to the student

1.4 Stakeholders

Provide a list of key stake-holders.

1. Students
2. Teacher Assistants
3. Faculty
4. Mumtahir(AIE)

Chapter 2

Project Description

2.1 Scope

The main scope that our product encompasses is to automate the whole process of demo evaluation. This includes checking submission, marking assignments, asking AI generated questions based on the assignment, taking quizzes, evaluating demos and finally, generating a report of the whole evaluation. This means that not only the fatigue of the Teaching Assistant would be reduced, but also, an AI based demo conduction would ensure non-biasness and thus, a fair marking for every student.

2.2 Modules

Given the current scope, our product aims to implement the following modules:

2.2.1 Module 1

The first Module serves to implement the foundation of our product. A strong foundation would lead to an efficient workflow. Thus, offering the following features: (Client sideWeb Application)

1. Figma Workflow
2. Initial Usecases (Login, Register, Submission Download and Facial Detection)

2.2.2 Module 2

With the basic foundation complete, our next objective is to start implementing the necessary functionalities. The second module would therefore offer the following features: (Client and Admin Side Web Application)

1. Assignment Plagiarism
2. Speech To Text Conversion
3. Text to Speech Conversion
4. Question generation based on Assignment
5. Answer Verification of the student by the AI model

2.2.3 Module 3

With the first two modules, the integral structure of our product is basically complete. Now we need to involve more important features that would improve the quality and credibility of our product. Thus, for our third module, we aim to implement the following features: (Client and Admin Side Web Application)

1. Facial Expression Evaluation to improve the credibility of our demos
2. Generate the report of demos
3. Submission Checking

2.2.4 Module 4

For our last module, our product is basically complete. For any product, Testing is as important as the rest of the implementation, so finally, our final module would offer testing as well as the deployment of our product. (Client and Admin Side Web Application)

1. Deployment
2. Testing
3. CICD

2.3 Tools and Technologies

- Git
- PyTorch
- Python
- MongoDB
- Flask
- Node Js
- GitHub

2.4 WorkDivision

For each module and respective feature, assign responsibility to a team member

Aaimlik Abbasi	Affan Anwar	Muhammad Hammad
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SRS Document Use cases Assignment plagiarism Speech to text Generate report Deployment	SRS Document Use cases Text to speech Question generation based on assignment Testing Integration	Poster Figma Answer verification Facial expression recognition Submission Checking
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2.5 TimeLine

2.5.1 Mid Evaluation

- SRSdocument
- Poster
- Figma (workflow)
- Use Case

2.5.2 Final(FYP-1)

- Assignment plagiarism
- Speech to text
- Text to speech
- Question generation based on assignment
- Answer verification

2.5.3 Pre-job fair evaluation

- Facial expression evaluation
- Generate report
- Submission check
- Integration

2.5.4 Final(FYP-2)

- Deployment
- Testing
- CICD

3. SRS

We are developing tools to implement a standardized evaluation system. This system will incorporate live demo sessions, assignment assessments, and evaluations. Additionally, it will utilize facial recognition technology to enhance the integrity of the process. Our goal is to ensure consistent evaluation criteria across all assessments. This approach aims to provide a fair and reliable evaluation experience for students and educators.

3.1 Lists of Features

1. Facial recognition
2. Plagiarism detection
3. Slot management
4. Submission checking
5. Generate report
6. Evaluate assignment
7. Slot management

3.2 Functional Requirement

- System will enable the student to login based on facial recognition.
- System will enable the student to be able to select their preferred time slots for Live demo according to their own convenience.
- System shall conduct the live demo, student's speech will be converted into text For quizzes and assessments.
- System will download student submissions and evaluate them.
- System will apply an algorithm to check submitted code for plagiarism
- System will generate the comprehensive report ,thus summarizing the results
- System shall enable the teacher to enroll student and release slot

3.3 Quality Attribute

Scalability

The system can handle an increasing number of the users.

Reliability

The system continues to perform its function without failure over time.

Usability

The system shall be easy to use.

Availability

The system shall enable the time when and where and to whom the system will be Available.

Secure

The personal information of students will be secure in the database

Maintainability

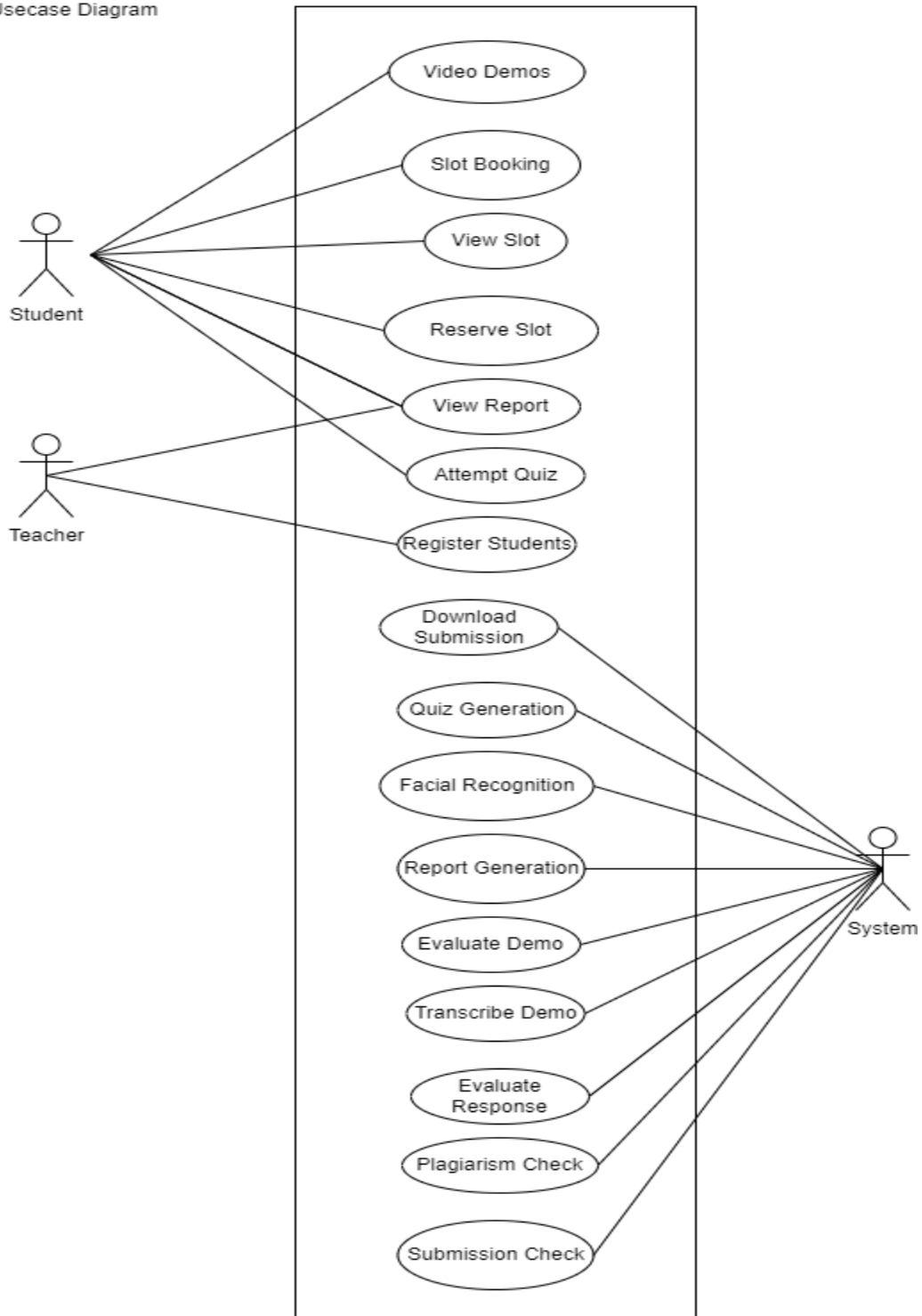
The system will be developed in a modular fashion, making it easy to update and modify .

3.4 Nonfunctional requirement

- The user will be able to login within 5 seconds after providing the right credential.
- The system must verify the credential with the database before logging in the user.
- The system should be easy to use, as its interface should be intuitive.
- The system should regularly backup the data and allow the quick recovery in case of the loss
- The user should be able to enter the meeting within 10 second of the meeting

4. Use Case Diagram:

Usecase Diagram



Mumtahir

Figure 1

5.High-Level Use Case

5.1:Video demos

Use Case	Video demos
Actors	Student
Type	primary
Description	The student will join the meeting at the allocated time and complete the quiz.

5.2 Slot Booking

Use Case	Slot Booking
Actors	Student
Type	primary
Description	The student select the available timeslot for demo and confirm the booking

5.3 Download Submission

Use Case	Download Submission
Actors	Student,teacher
Type	primary
Description	The student can download their submitted assignment for review or personal records

5.4 Facial Recognition

Use Case	Facial Recognition
Actors	system
Type	Secondary
Description	The system performs facial recognition to verify the identity of the student during the assignment demo

5.5 Generate Report

Use Case	Generate Report
Actors	system
Type	Secondary
Description	The system generates a report summarizing the student's assignment demo performance

5.6 Plagiarism Check

Use Case	Plagiarism Check
Actors	system
Type	Secondary
Description	The system performs a plagiarism check on the submitted assignment by comparing its content against a database of existing documents and online sources .

6. Extended Use Case

6.1 Video Demos

Use Case Name	Video Demos
Scope	Mumtahir
Level	User Goal
Primary Actor	Student
Stakeholders and Interest	Student: Wants to successfully demonstrate their knowledge System: Maintaining the integrity of evaluation process
Precondition	Student must have internet connection Student must log in to the system
PostCondition	The student has completed the demo, and the performance data is recorded for evaluation.
Main Success Scenario	

	Actors action	System responsibility
	1. Student login system and join the schedule meeting 2. Student answer the quiz question	1. System verify the student id through facial recognition 2. The system records the demo and collect performance
Extensions	1a. if facial recognition fail ,system prompts the student to retry 2a. If the student fails to join on time, the demo is canceled	
Special Requirement	1. Stable internet connection 2. Simple and easy UI 3. The user has been verified.	

6.2 Slot Booking

Use Case Name	Slot booking	
Scope	Mumtahir	
Level	User Goal	
Primary Actor	Student	
Stakeholders and Interest	Student: Want a convenient time for their demo Teacher : Aims a fair evaluation process System: Needs to maintain an organized schedule to prevent conflicts	
Precondition	Student must login Student should have access to available time slot	
PostCondition	The time slot is successfully booked for the student	
Main Success Scenario	Actors action	System responsibility
	1. student navigates to the slot booking section 2. The student selects a preferred time slot	1. The system checks the availability of the selected time slot 2. The system confirms the booking and updates the schedule.

Extensions	1a. If the selected time slot is unavailable, the system prompts the student to choose another time.
Special Requirement	1. The system must ensure that time slots are updated 2. Stable internet connection

6.3 Submission Download

Use Case Name	Submission Download	
Scope	Mumtahir	
Level	System goal level	
Primary Actor	System	
Stakeholders and Interest	System: Facilitates downloading of the submissions securely and efficiently.	
Precondition	Submissions must be available in the system.	
PostCondition	The submission is successfully downloaded to the teacher's local system.	
Main Success Scenario		
	Actors action	System responsibility
	1. The teacher initiates the download request.	1. The system retrieves the requested submission from the database.
Extensions	1a If the submission is unavailable, the system displays an error message 2b If a file download fails due to network issues, the system allows retries.	
Special Requirement	1. The downloaded file should be in its original format without modification 2. The system must support secure file transmission	

6.4 Facial Recognition

Use Case Name	Facial Recognition	
Scope	Mumtahir	
Level	System goal level	
Primary Actor	System	
Stakeholders and Interest	System: Must accurately perform facial recognition to ensure the correct identification of users.	
Precondition	Student must be present in front of the camera Student must have being log in	
PostCondition	The system successfully identified the student	
Main Success Scenario		
	Actors action	System responsibility
	1. The student sits in front of the camera	1. The system captures the student's facial image using the camera. 2. If the identity is confirmed, the system allows the student to proceed with the demo.
Extensions	1a. If the camera is unable to capture a clear image, the system alerts the student to adjust their position. 2a. If the system fails to recognize the student ,demo are cancel	
Special Requirement	1. The facial recognition system must operate in various lighting conditions	

6.5 Generate Report

Use Case Name	Generate Report	
Scope	Mumtahir	
Level	System goal level	
Primary Actor	System	
Stakeholders and Interest	System: Must accurately generate reports based on data	

	collected during the assignment demos Student: Student can access their performance				
Precondition	The assignment demo must have been completed The system must have gathered all necessary data				
PostCondition	A detailed report is generated and made accessible to both the teacher and the student				
Main Success Scenario	<table border="1"> <thead> <tr> <th>Actors action</th><th>System responsibility</th></tr> </thead> <tbody> <tr> <td>The system detects that the assignment demo has concluded.</td><td>The system generates a comprehensive report summarizing the student's performance.</td></tr> </tbody> </table>	Actors action	System responsibility	The system detects that the assignment demo has concluded.	The system generates a comprehensive report summarizing the student's performance.
Actors action	System responsibility				
The system detects that the assignment demo has concluded.	The system generates a comprehensive report summarizing the student's performance.				
Extensions	If the report fails to generate, the system will log the error and inform the teacher				
Special Requirement	The report should follow a standard format to keep all reports consistent.				

6.6 Plagiarism Check

Use Case Name	Plagiarism check				
Scope	Mumtahir				
Level	System goal level				
Primary Actor	System				
Stakeholders and Interest	Teacher: Ensures fairness in student submissions.				
Precondition	The student has submitted an assignment for evaluation.				
PostCondition	The system has analyzed the assignment for plagiarism and generated a report.				
Main Success Scenario	<table border="1"> <thead> <tr> <th>Actors action</th><th>System responsibility</th></tr> </thead> <tbody> <tr> <td>The student submits their assignment. The system initiates the plagiarism check.</td><td>Analyze the submitted assignment against a database of existing work. Generate a plagiarism report</td></tr> </tbody> </table>	Actors action	System responsibility	The student submits their assignment. The system initiates the plagiarism check.	Analyze the submitted assignment against a database of existing work. Generate a plagiarism report
Actors action	System responsibility				
The student submits their assignment. The system initiates the plagiarism check.	Analyze the submitted assignment against a database of existing work. Generate a plagiarism report				
Extensions	1a. If the submission file format is incorrect, the system tell the				

	student and award him zero
Special Requirement	The plagiarism check should provide a percentage indicating the level of originality

7. Domain model:

Domain Model

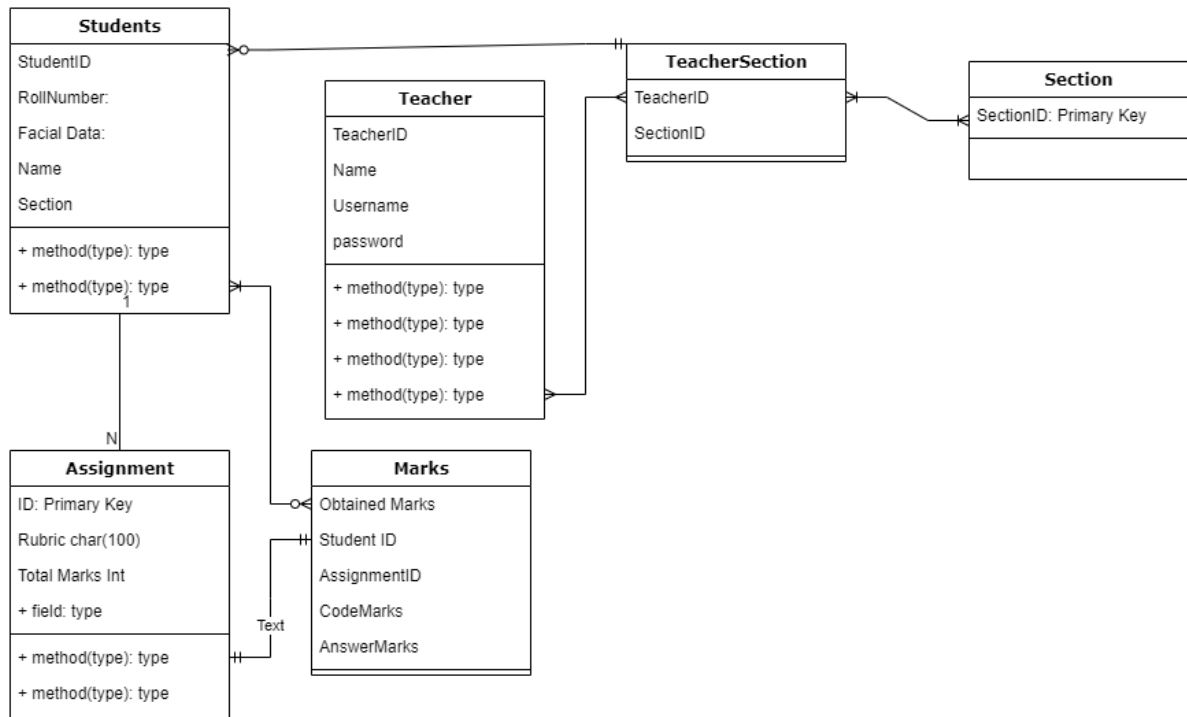


Figure 2

8. Sequence Diagram – SSD

8.1.Video Demo

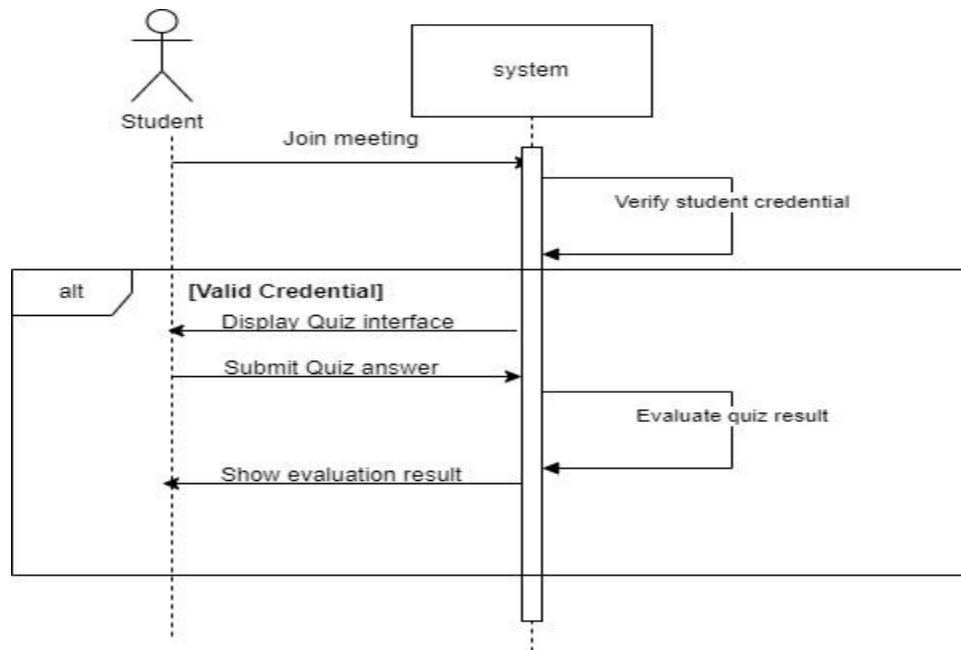


Figure 3

8.2 Check Marks

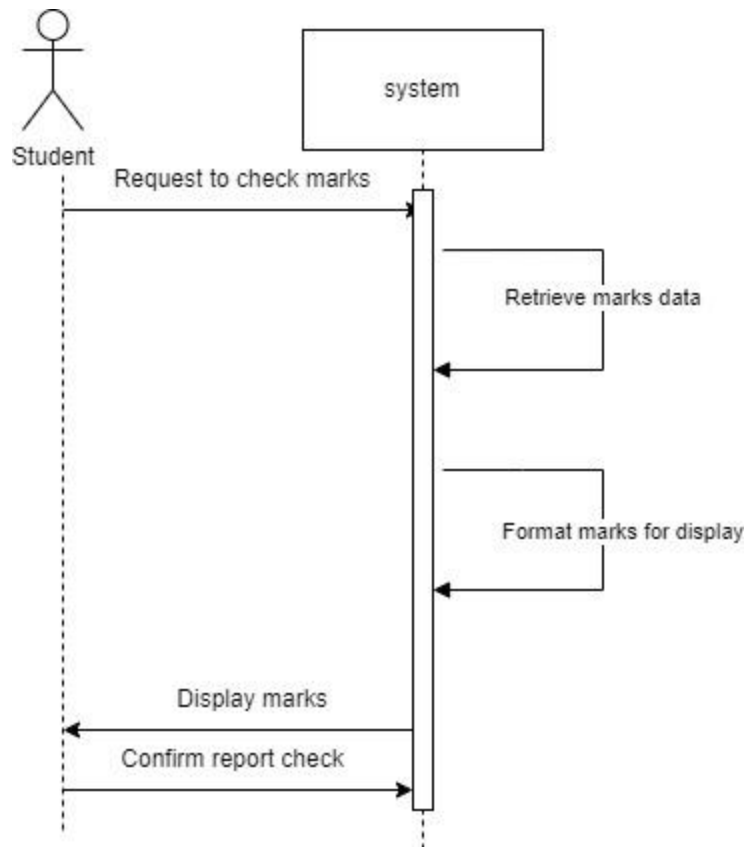


Figure 4

8.3 Download report

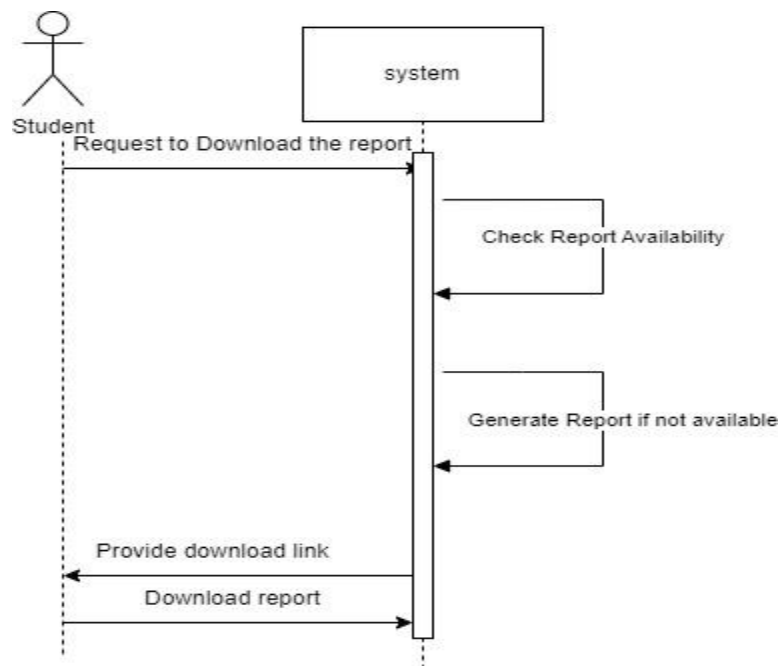


Figure 5

8.4 View report

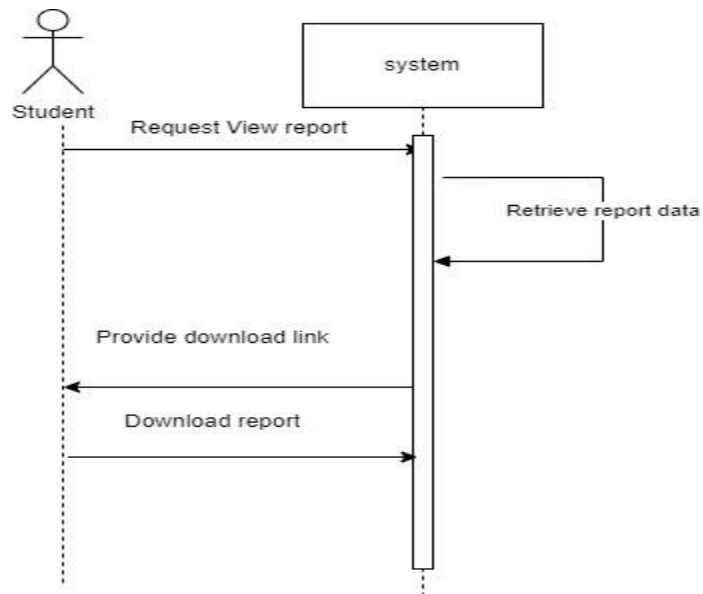


Figure 6

8.5 Update Slots:

Update Slots

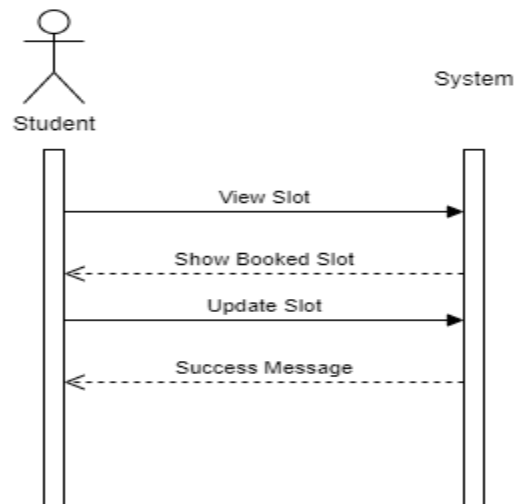


Figure 7

8.6 Enroll Students:

Enroll Students

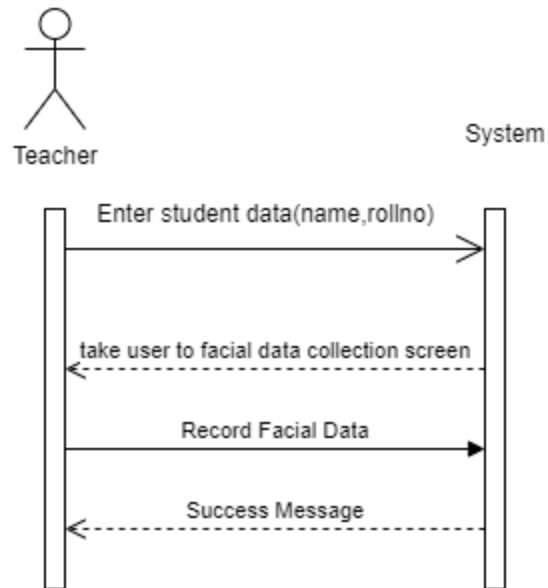


Figure 8

8.7 Evaluating Demo:

Evaluating Demo

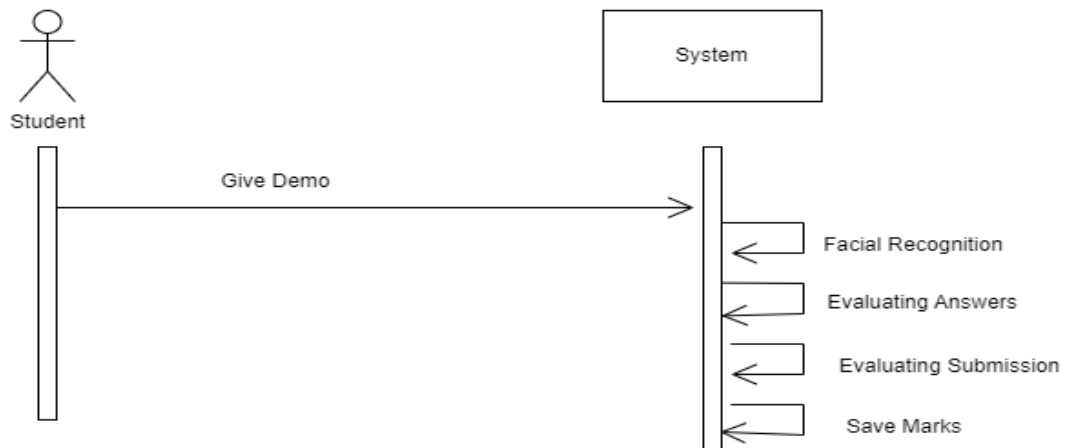


Figure 9

8.8 Release Slots:

Release Slots

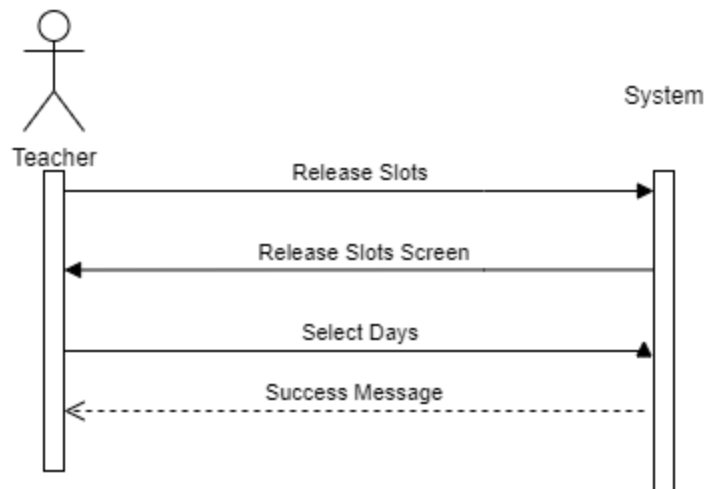


Figure 10

9. Activity Diagram/ Data Flow Diagram

9.1 Video Demo

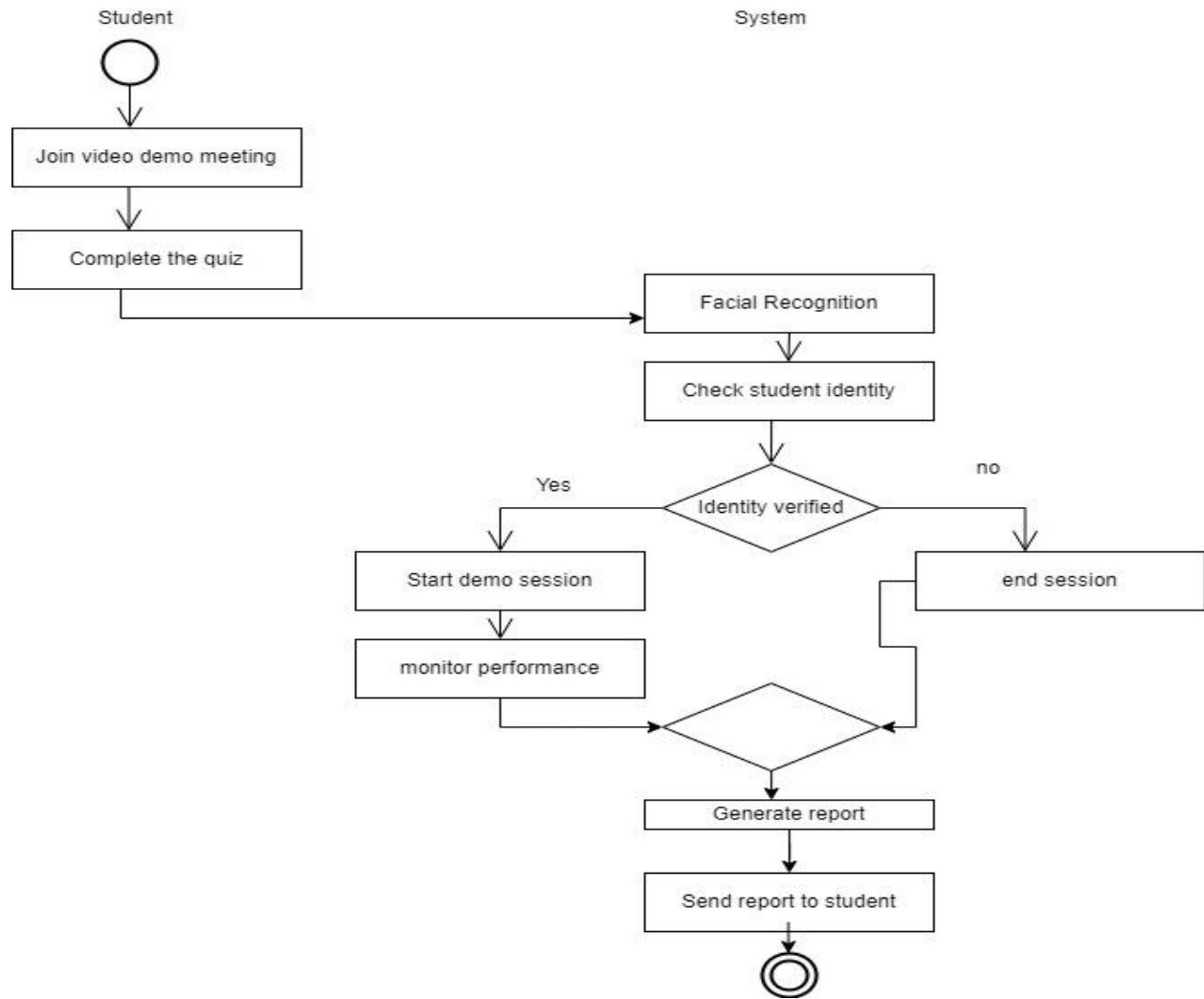


Figure 11

9.2 Check marks

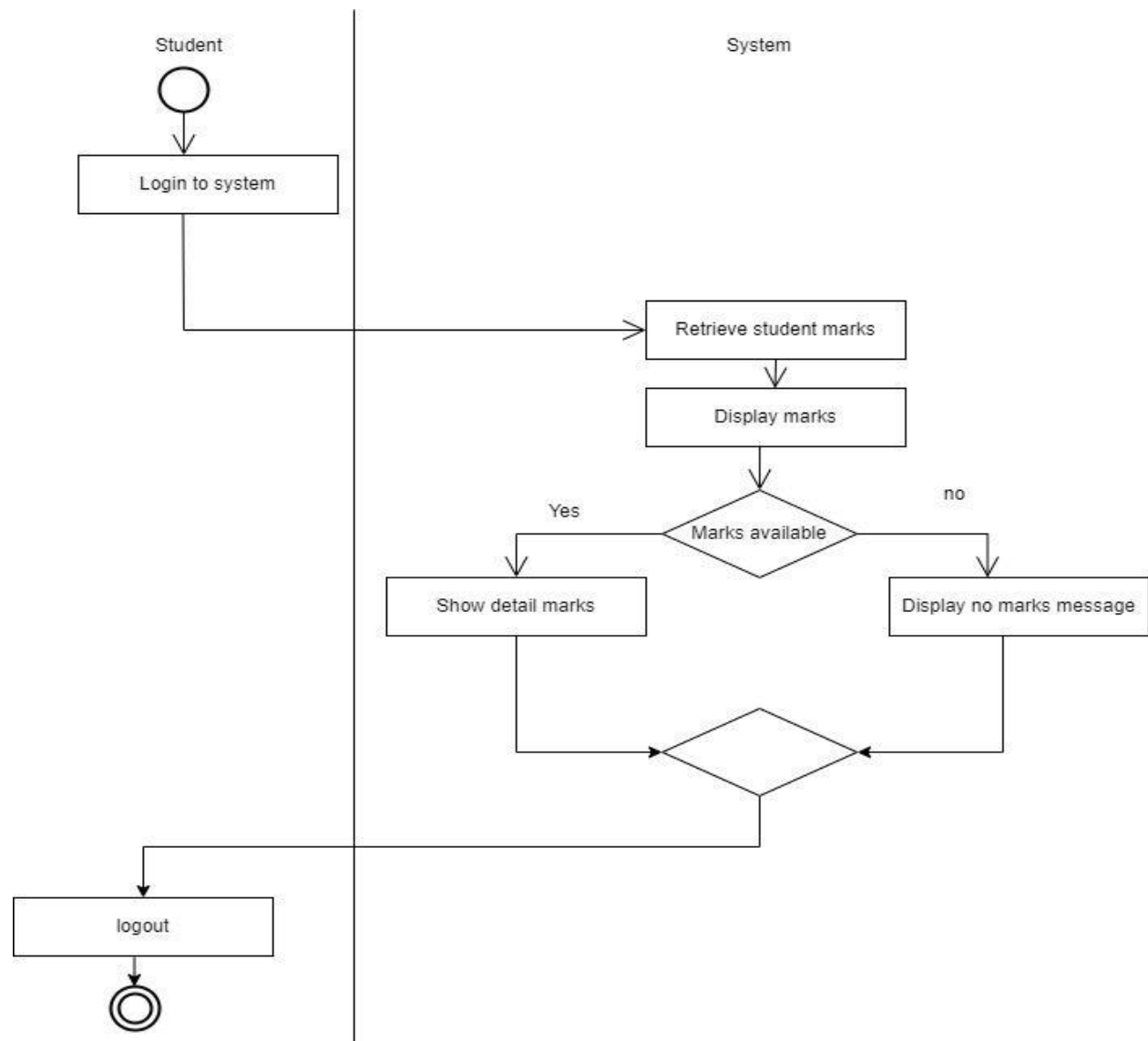


Figure 12

9.3 Download report

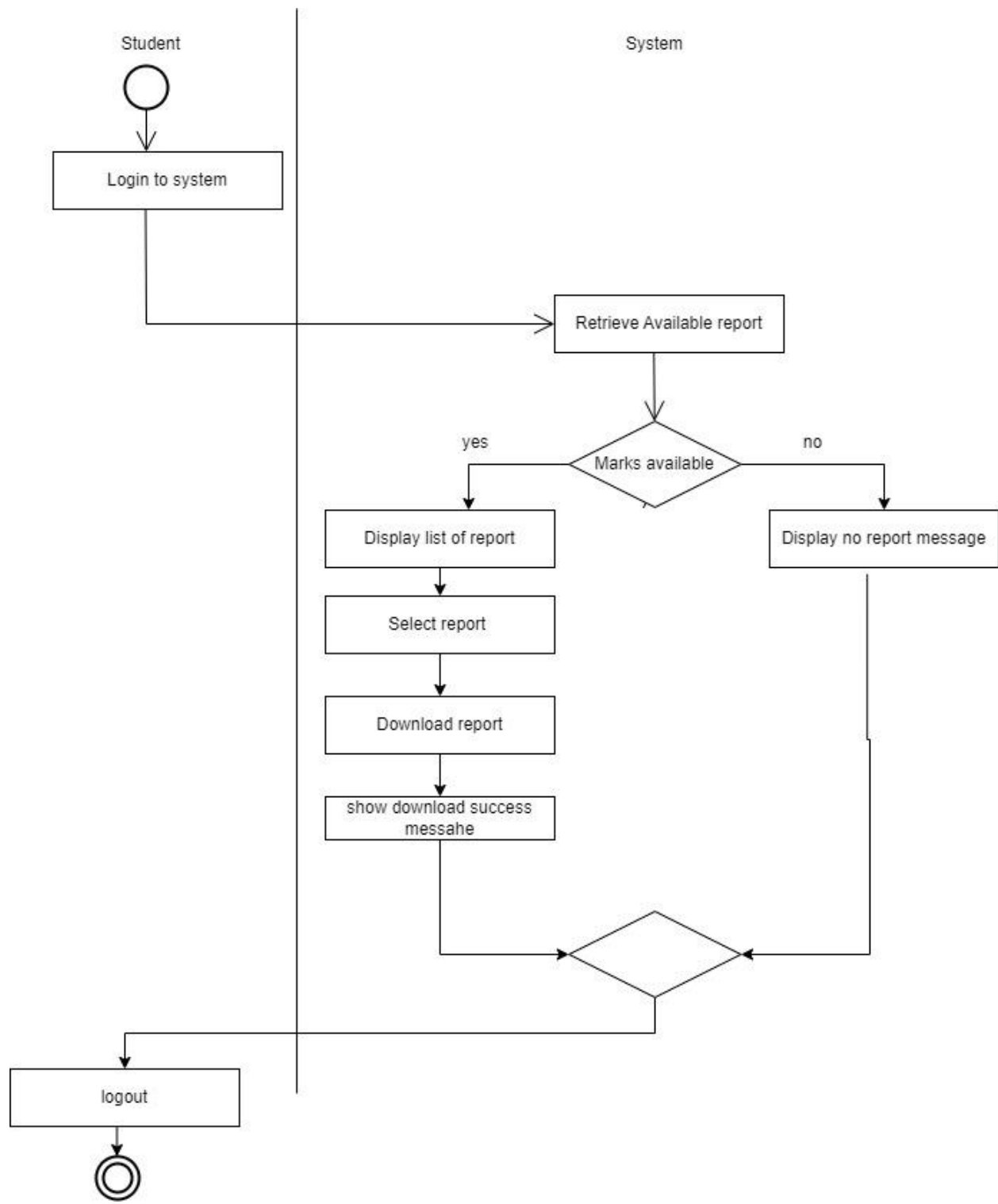


Figure 13

9.4 View report

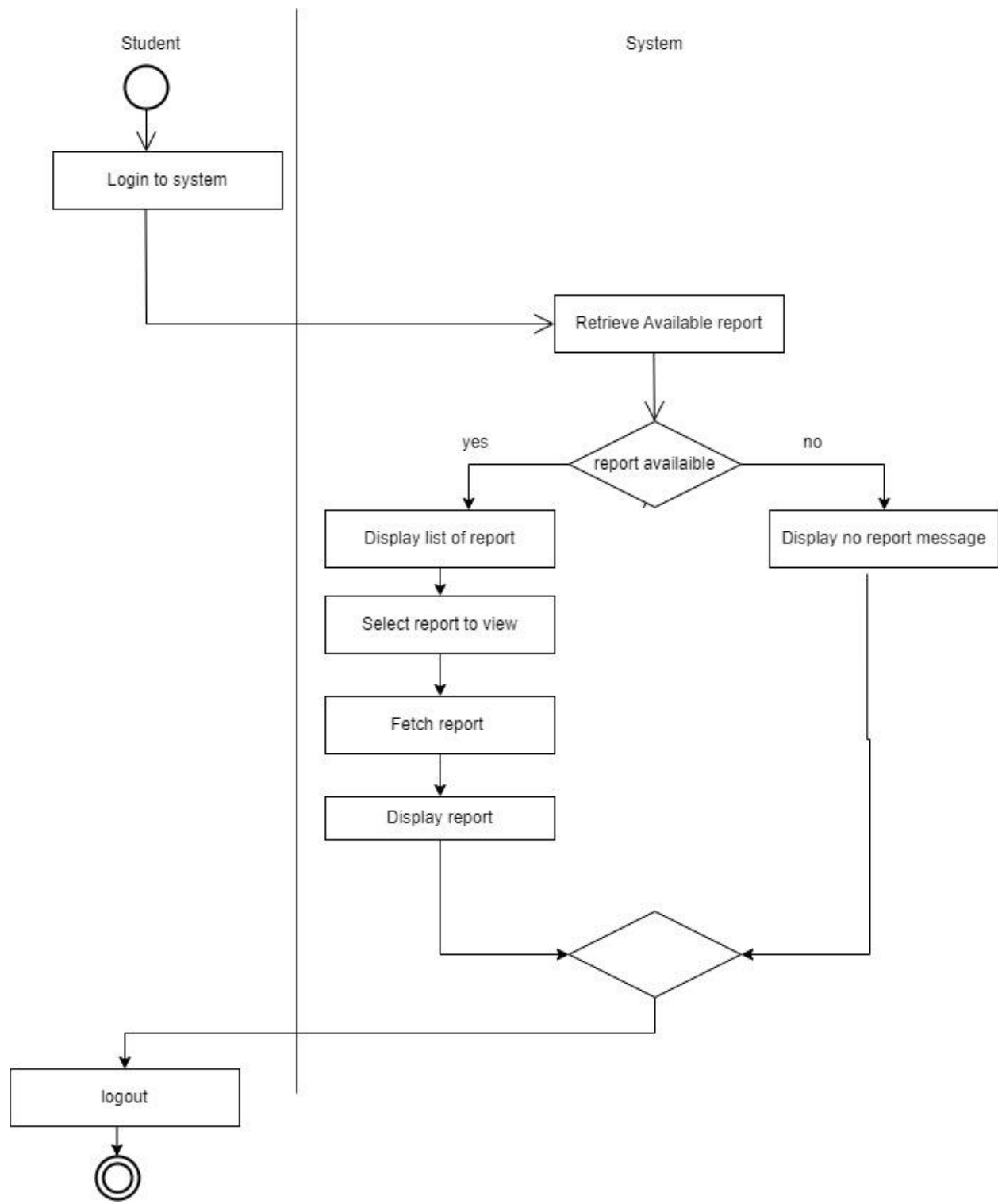


Figure 14

9.5 Evaluate Demo:

Evaluate Demo

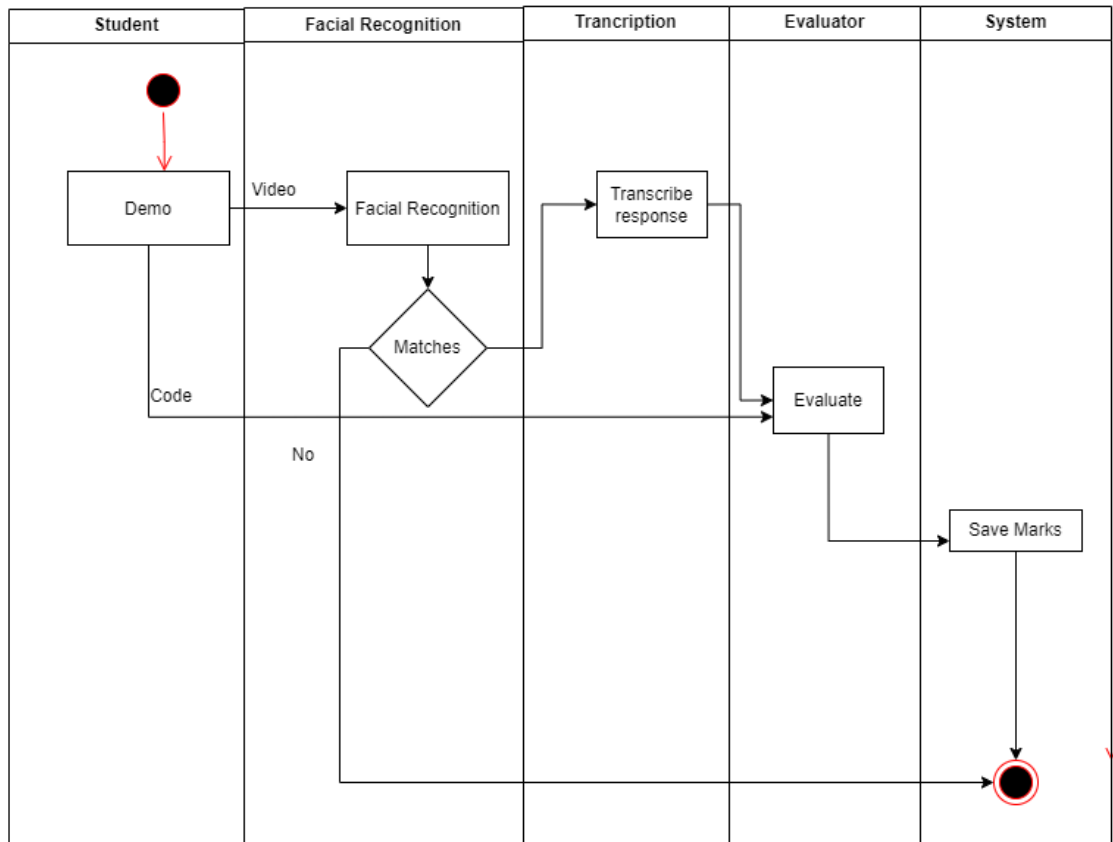


Figure 15

9.6 Enroll Students:

Enroll Students

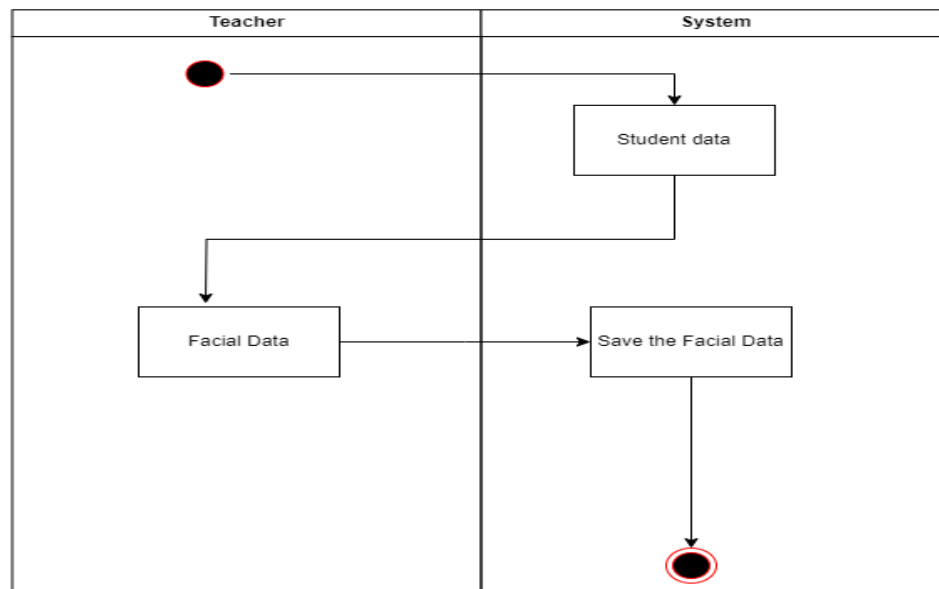


Figure 16

9.7: Update Slots:

Update Slots

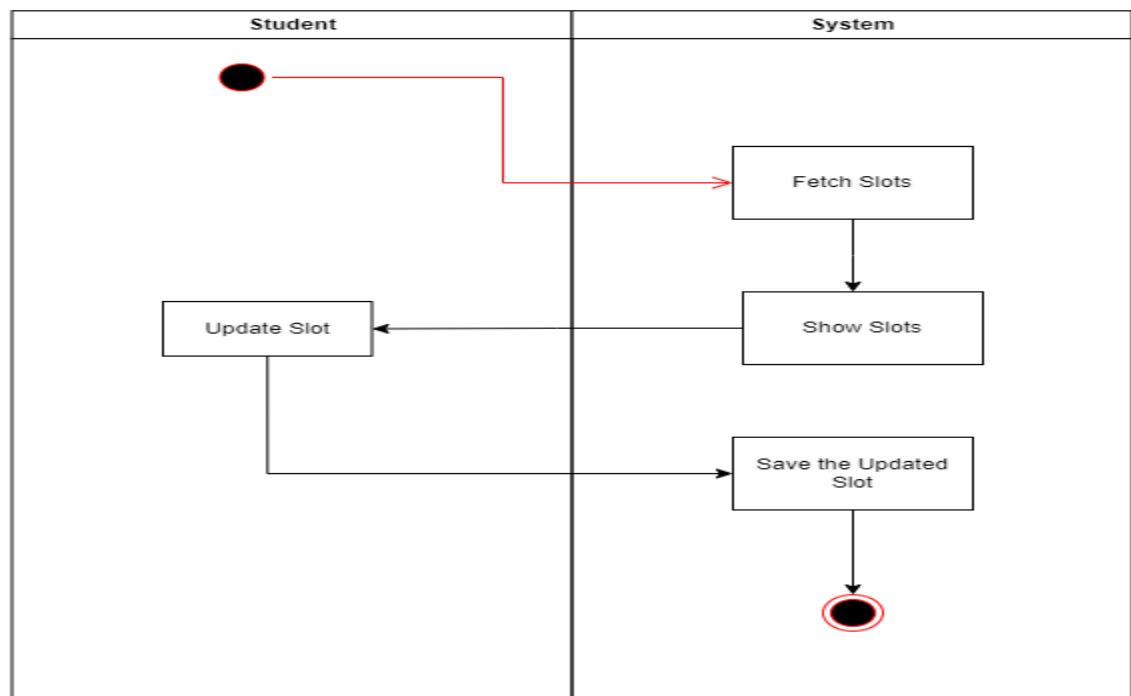


Figure 17

9.8 Release Slots:

Release Slots

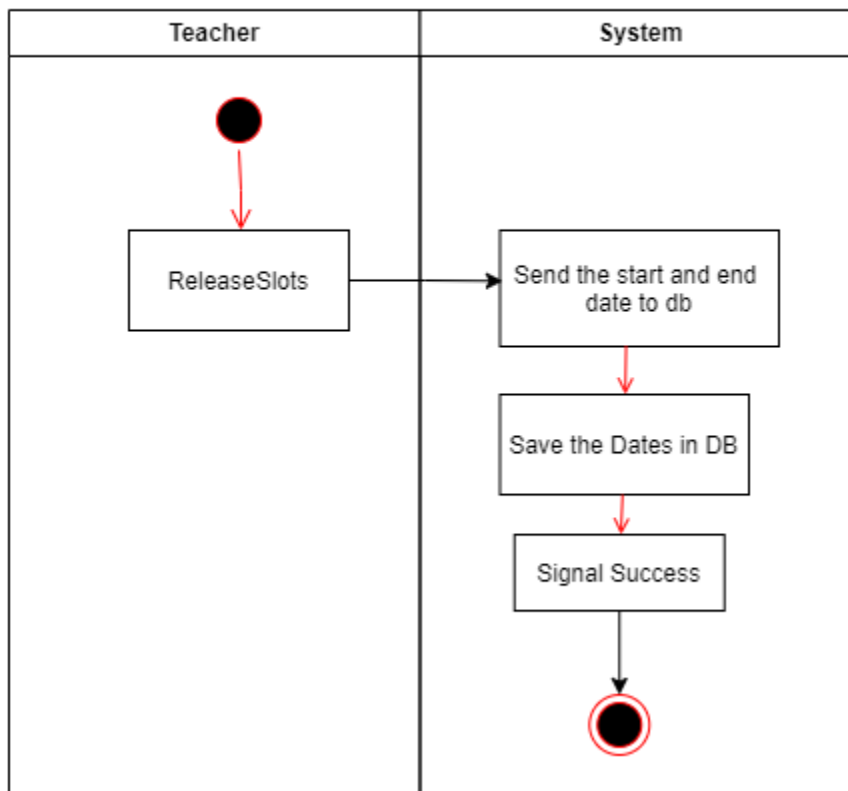


Figure 18

10. Operations Contracts:

10.1 Video Demo

Operation: Video_demo()

Cross-reference: Use case to have video demo

Precondition:

- Student must have internet connection
- Student must log in to the system

Postcondition:

The student has completed the demo and the performance of data is recorded for evaluation

10.2 Slot Booking

Operation: Slot_booking()

Cross-reference: Use case to book your slot

Precondition:

- Student must login
- Student should have access to available time slot

Postcondition:

The time slot is successfully booked for the student

10.3 Submission Download

Operation: Submission_download()

Cross-reference: Use case to download submission

Precondition:

- Submissions must be available in the system.

Postcondition:

The submission is successfully downloaded to the teacher's local system.

10.4 Facial Recognition

Operation: Facial_recognition()

Cross-reference: Use case to recognize student identity

Precondition:

- Student must be present in front of the camera
- Student must have being log in

Postcondition:

The system successfully identified the student

10.5 Generate Report

Operation: Generate_report()

Cross-reference: Use case generate the report

Precondition:

- The assignment demo must have been completed
- The system must have gathered all necessary data

Postcondition:

A detailed report is generated and made accessible to both the teacher and the student.

10.6 Plagiarism Check

Operation: Plagiarism_check()

Cross-reference: Use case to detect plagiarism

Precondition:

- The student has submitted an assignment for evaluation.

Postcondition:

The system has analyzed the assignment for plagiarism and generated a report

11. Class Diagram

Class Diagram

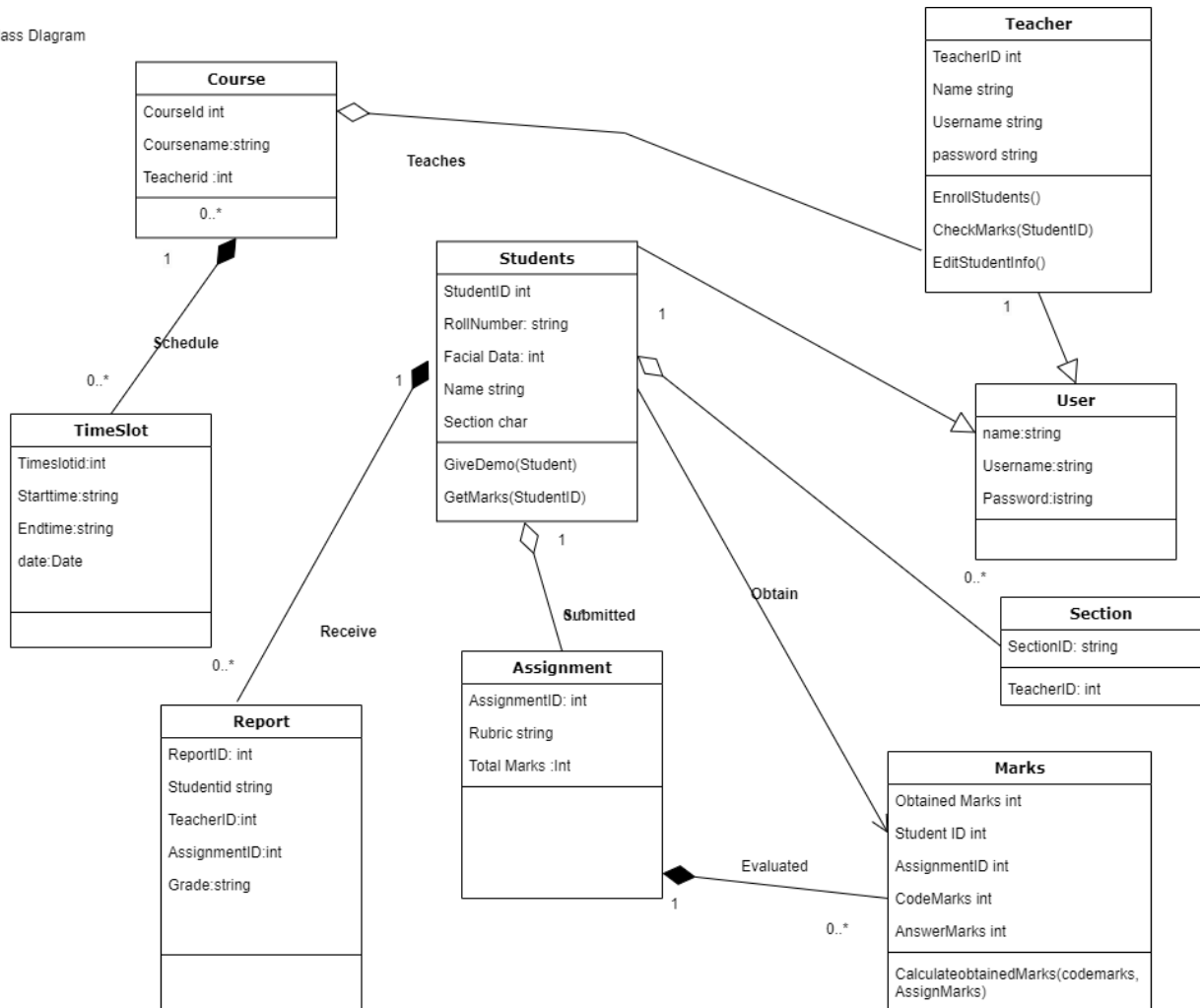


Figure 19

12. System Sequence Diagram

12.1 Video demo

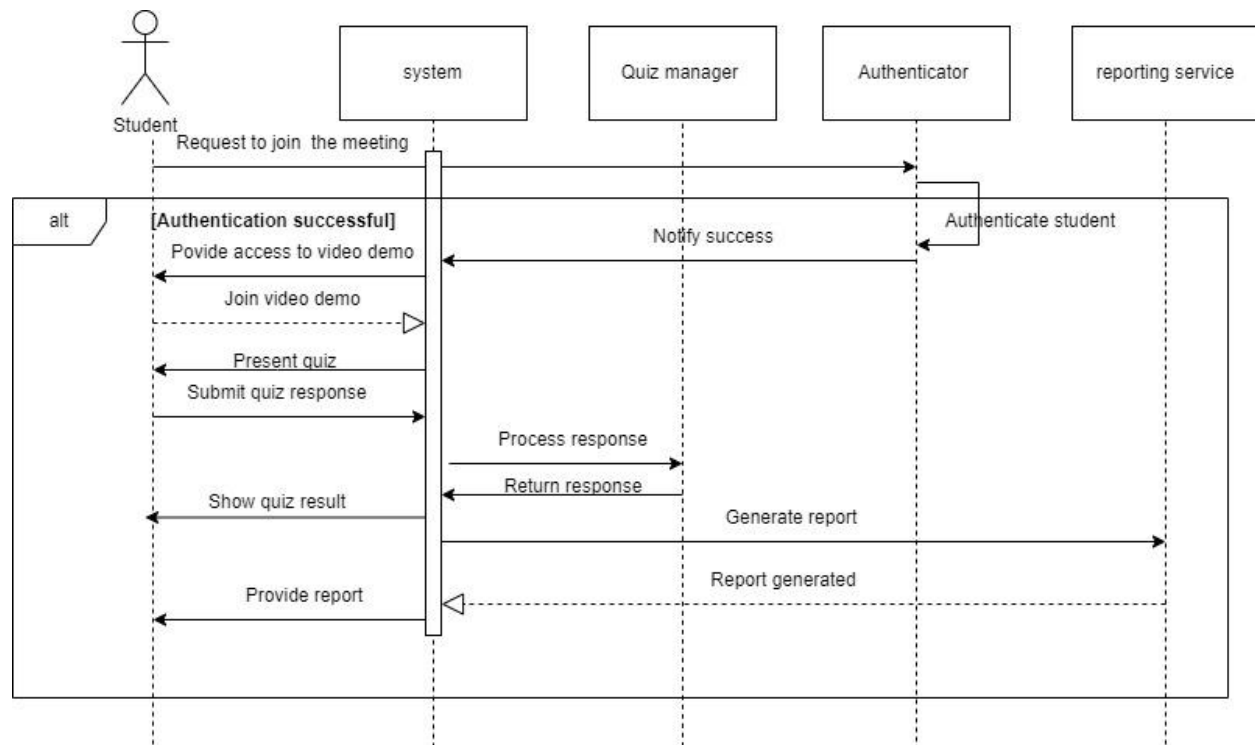


Figure 20

12.2 Check marks

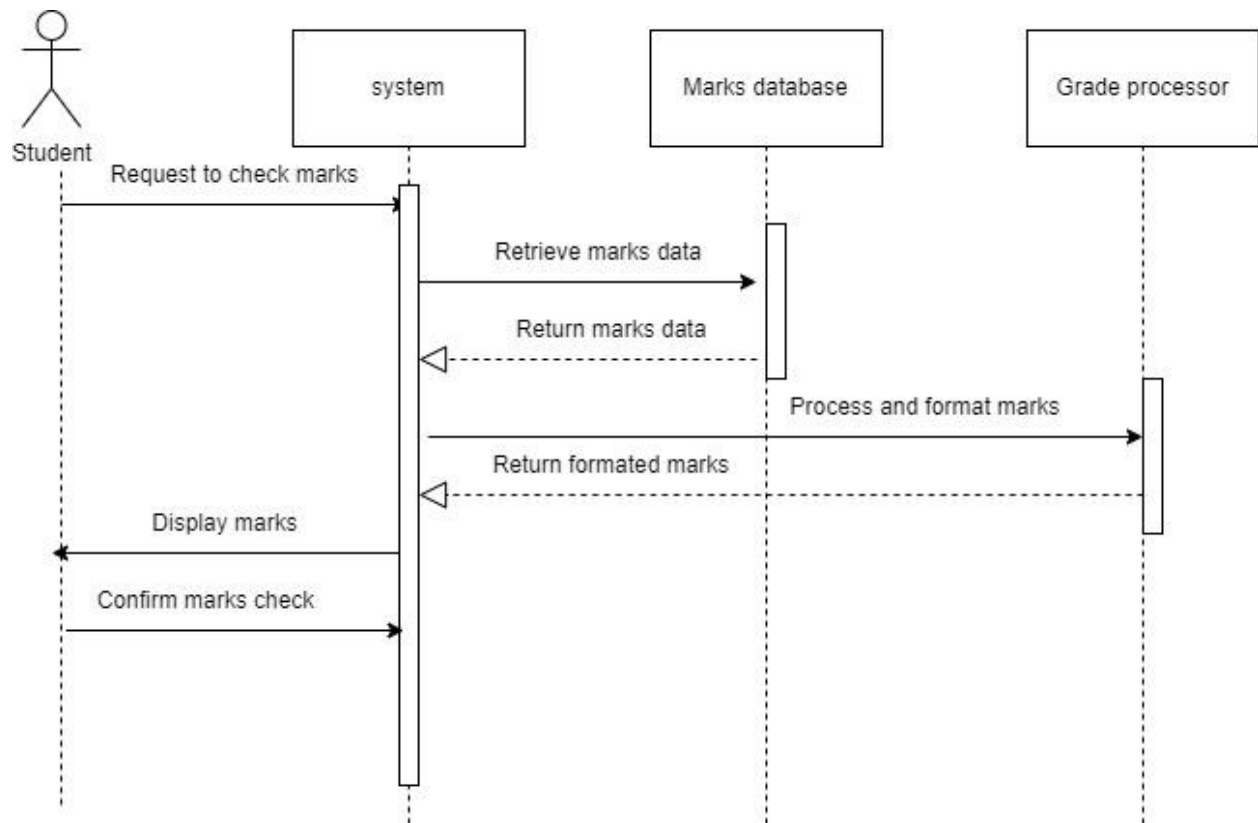


Figure 21

12.3.Download report

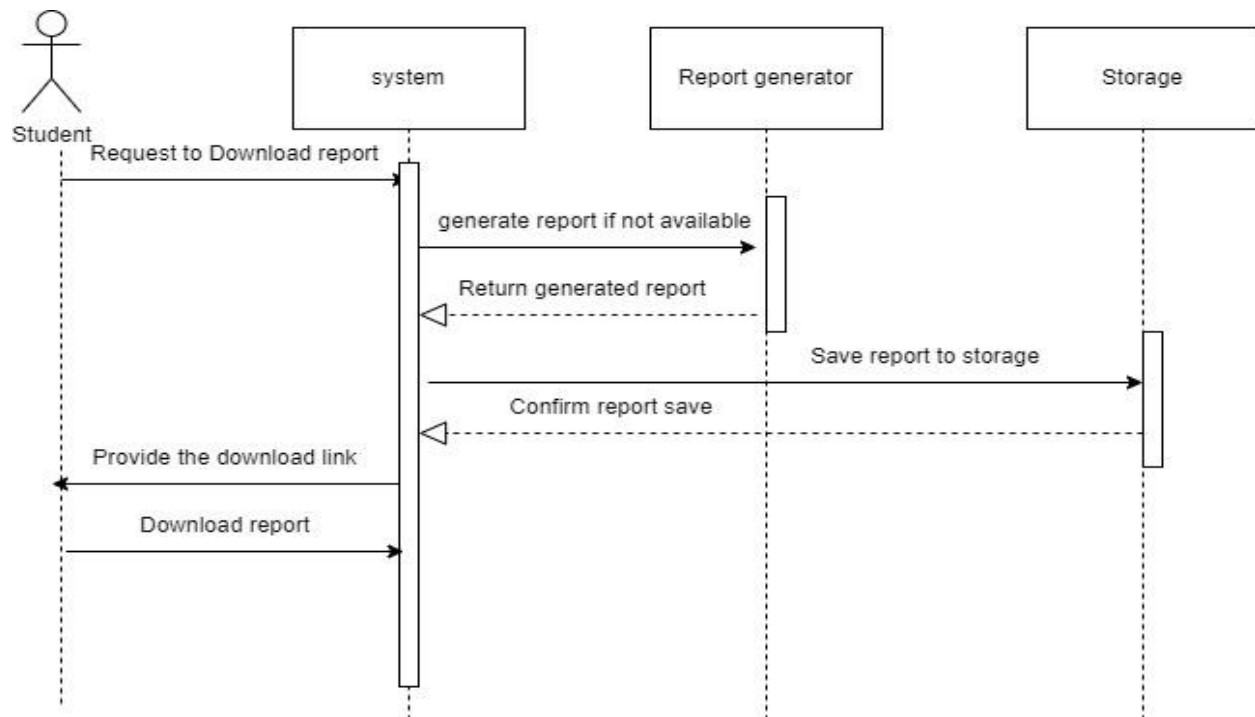


Figure 22

12.4 View report

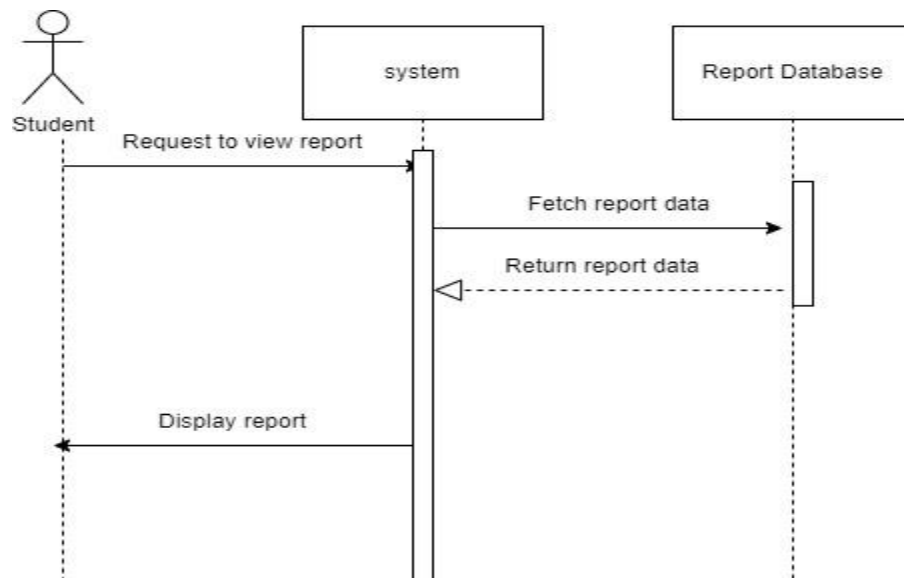


Figure 23

12.5 Evaluate Demo

Evaluate Demo

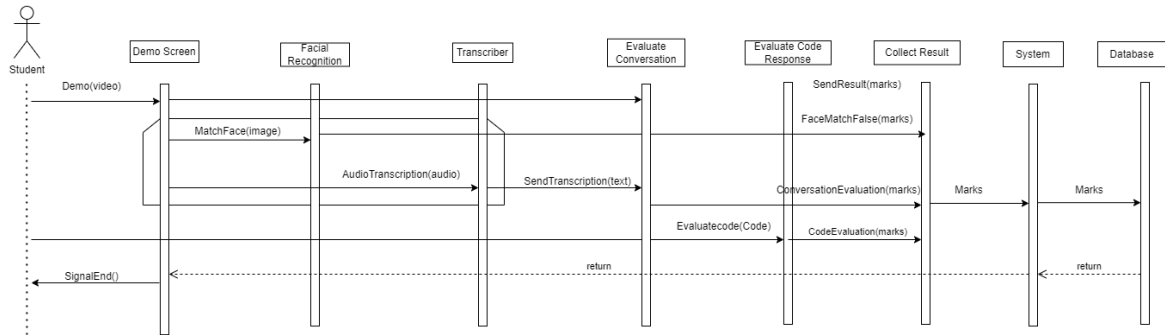


Figure 24

12.6 Enroll Students

Enroll Students

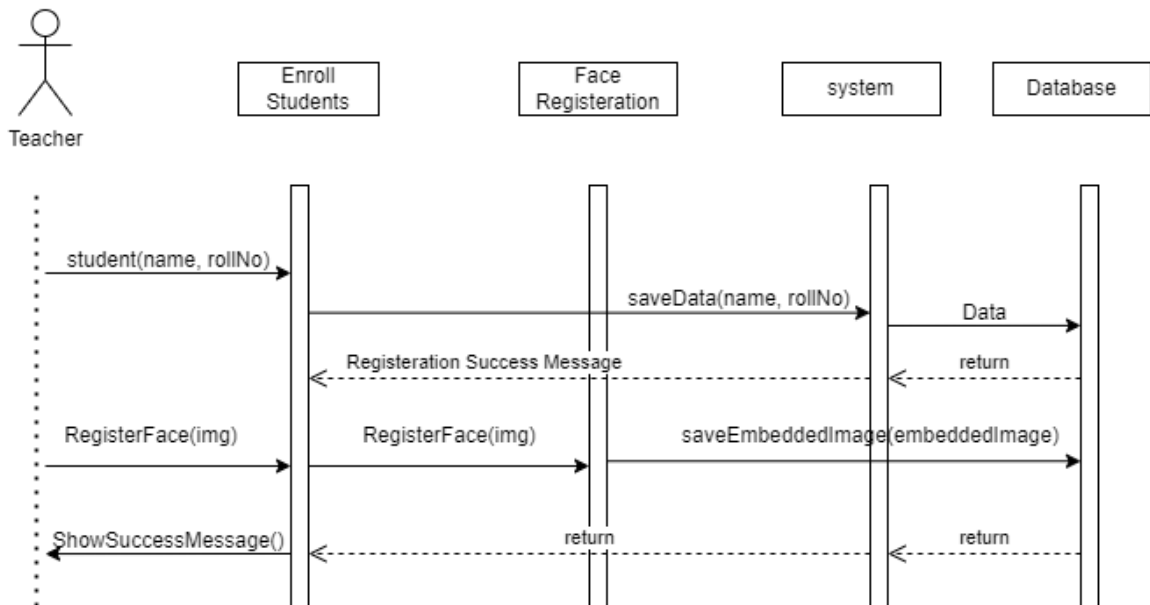


Figure 25

12.7 Release Slots

Release Slots

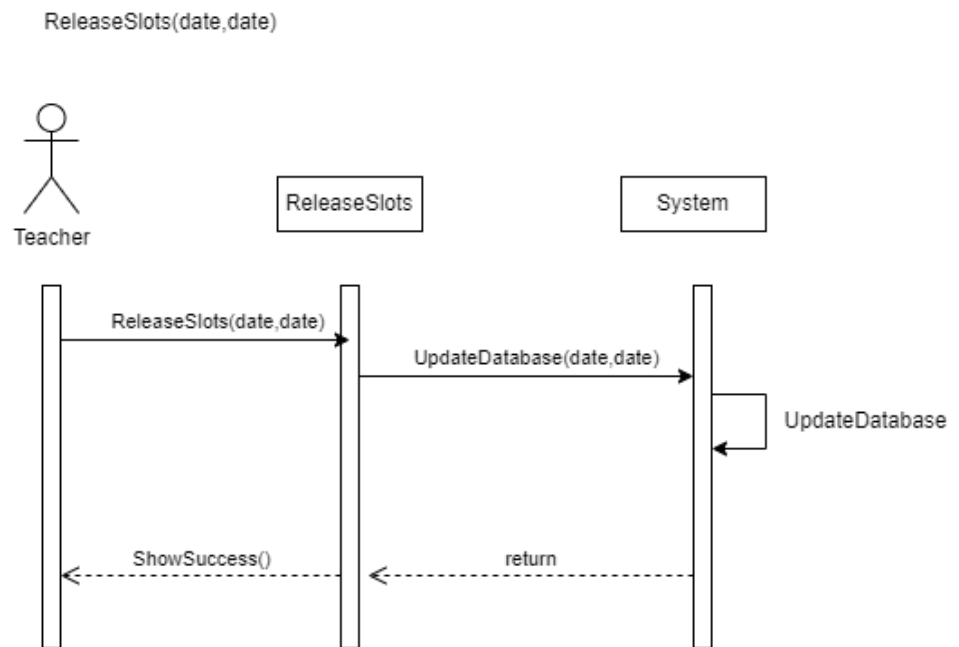


Figure 26

12.8 Update Slots:

Update Slots

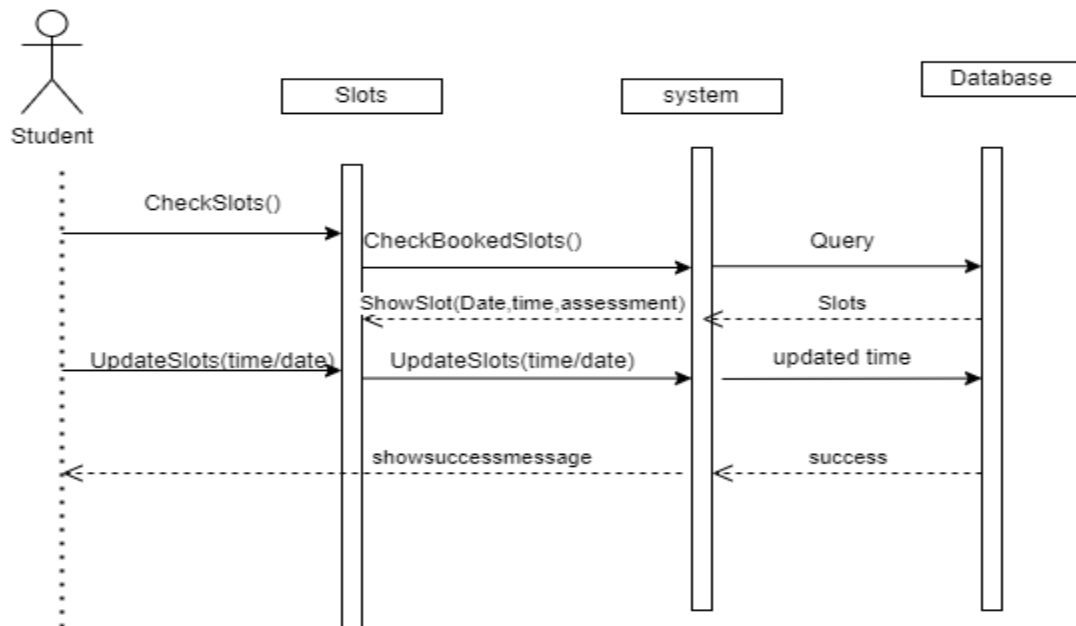


Figure 27

13 Json Schema(substituted for ERD):

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "title": "Mumtahir",
  "description": "Schema representing students, teachers, sections, assignments, and marks",
  "properties": {
    "students": {
      "type": "array",
      "description": "List of students",
      "items": {
        "type": "object",
        "properties": {
          "studentID": { "type": "integer" },
          "rollNumber": { "type": "string" },
          "facialData": { "type": "string" },
```

```

        "name": { "type": "string" },
        "section": {
            "type": "object",
            "properties": {
                "sectionID": { "type": "string" }
            },
            "required": ["sectionID"]
        }
    },
    "required": ["studentID", "rollNumber", "name", "section"]
}

},
"teachers": {
    "type": "array",
    "description": "List of teachers",
    "items": {
        "type": "object",
        "properties": {
            "teacherID": { "type": "integer" },
            "name": { "type": "string" },
            "username": { "type": "string" },
            "password": { "type": "string" }
        },
        "required": ["teacherID", "name", "username", "password"]
    }
},
"sections": {
    "type": "array",
    "description": "List of sections",
    "items": {
        "type": "object",
        "properties": {
            "sectionID": { "type": "string" }
        },
        "required": ["sectionID"]
    }
},
"teacherSections": {
    "type": "array",
    "description": "Mapping of teachers to sections",

```



```

    "items": {
      "type": "object",
      "properties": {
        "teacherID": { "type": "integer" },
        "sectionID": { "type": "string" }
      },
      "required": ["teacherID", "sectionID"]
    }
  },
  "assignments": {
    "type": "array",
    "description": "List of assignments",
    "items": {
      "type": "object",
      "properties": {
        "assignmentID": { "type": "integer" },
        "rubric": { "type": "string" },
        "totalMarks": { "type": "integer" }
      },
      "required": ["assignmentID", "rubric", "totalMarks"]
    }
  },
  "marks": {
    "type": "array",
    "description": "Marks obtained by students for assignments",
    "items": {
      "type": "object",
      "properties": {
        "obtainedMarks": { "type": "integer" },
        "studentID": { "type": "integer" },
        "assignmentID": { "type": "integer" },
        "codeMarks": { "type": "integer" },
        "answerMarks": { "type": "integer" }
      },
      "required": ["obtainedMarks", "studentID", "assignmentID",
"codeMarks", "answerMarks"]
    }
  }
},

```

```
"required": ["students", "teachers", "sections", "assignments",  
"marks"]  
}
```

14. Architecture Diagram

Architecture Diagram

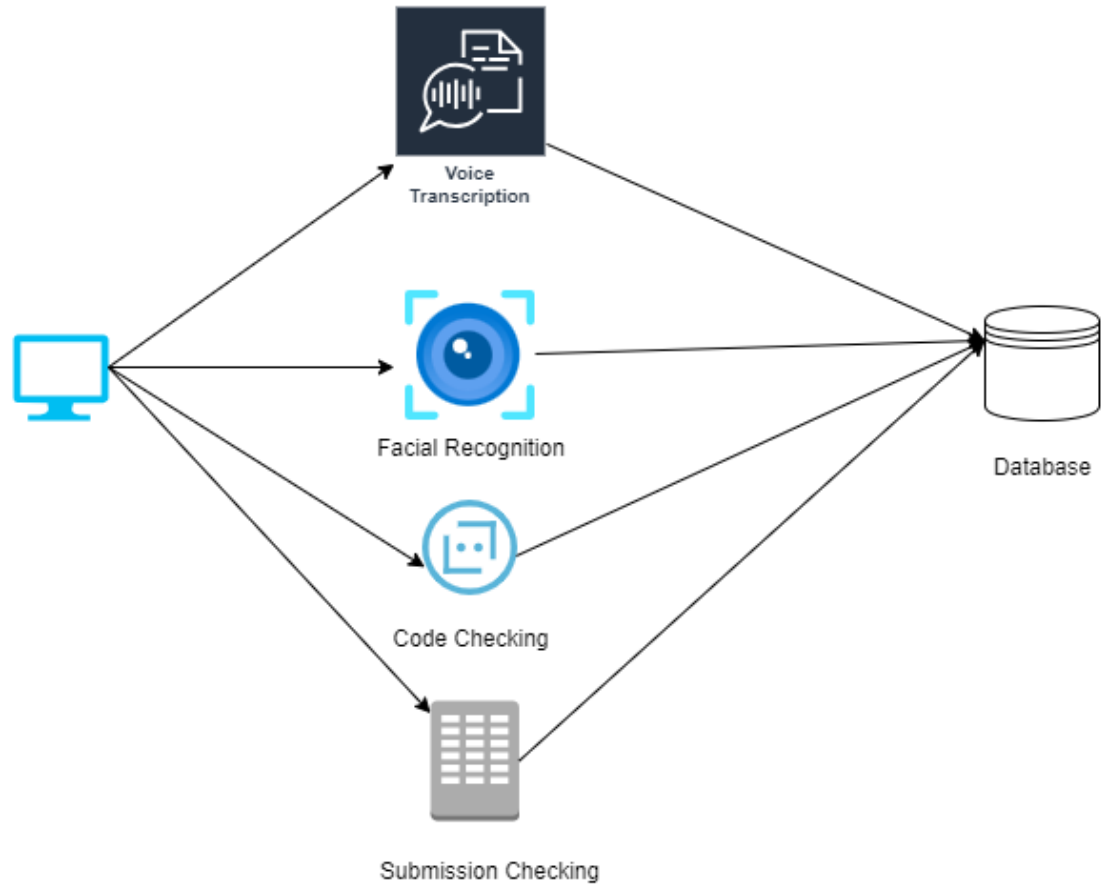


Figure 28

15. Box and Line Diagram:

Box and Line Diagram

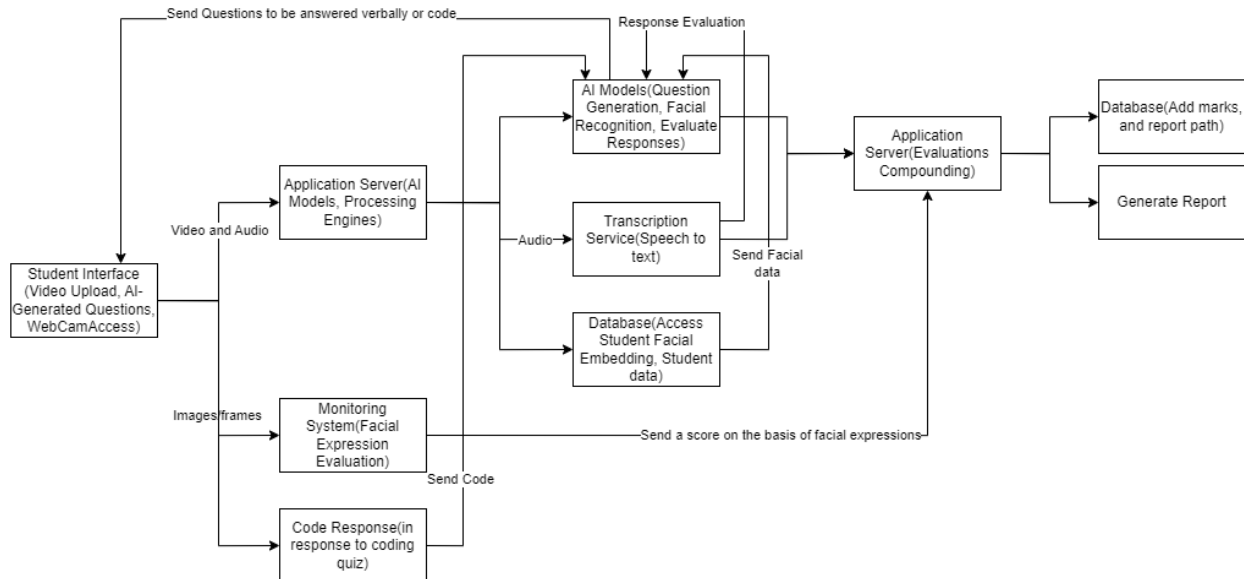


Figure 29

16. State Transition Diagram

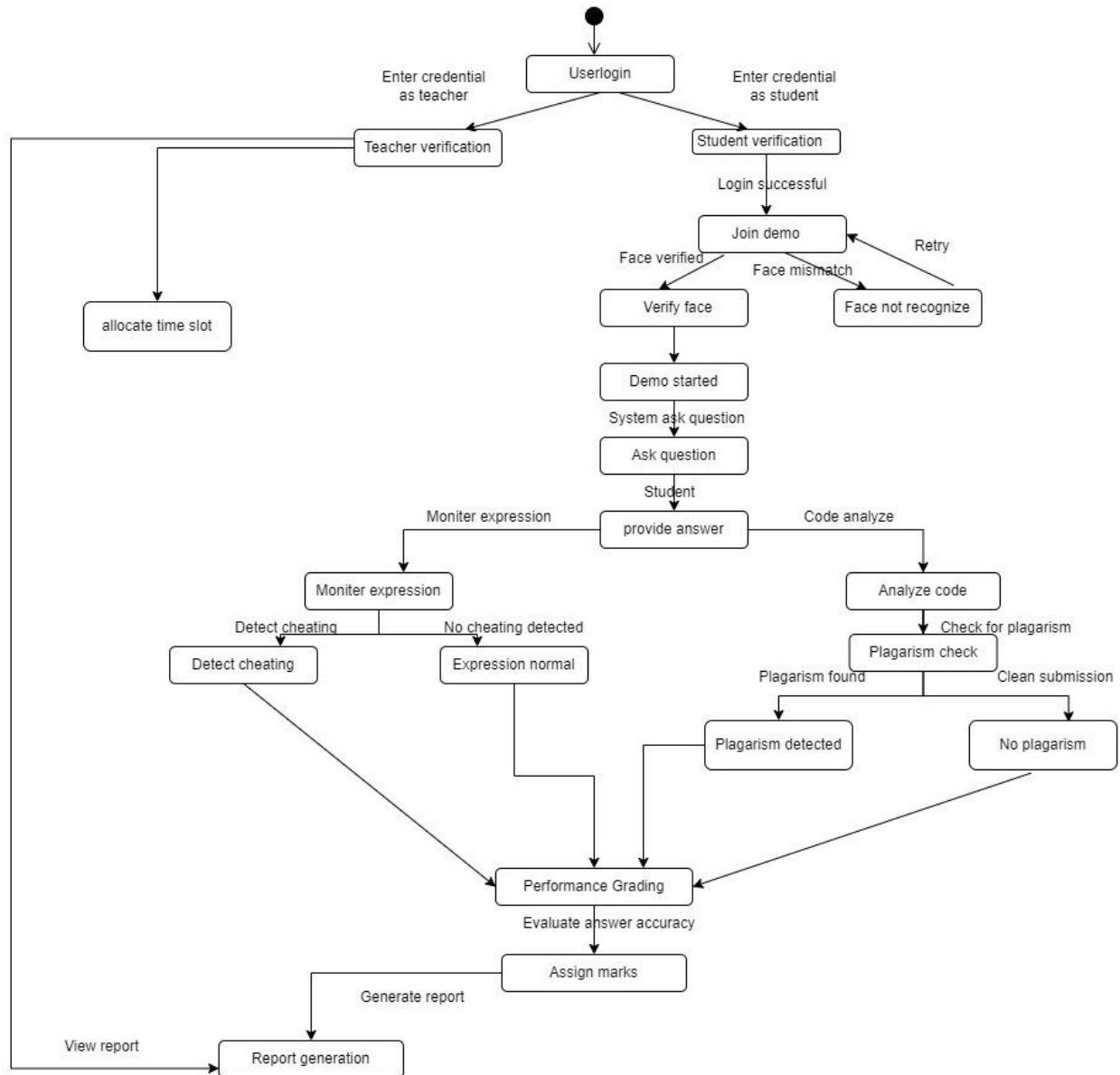


Figure 30

17. Implementation Details

17.1. Front-End Implementation

The front-end implementation for the desktop application is developed using HTML, CSS, and Bootstrap to ensure a responsive and visually appealing interface. It includes key features like a dashboard, slot booking, quiz interface, and report viewer, all styled for consistency across platforms. The interface integrates seamlessly with the back-end services via secure APIs, with a focus on usability and accessibility.

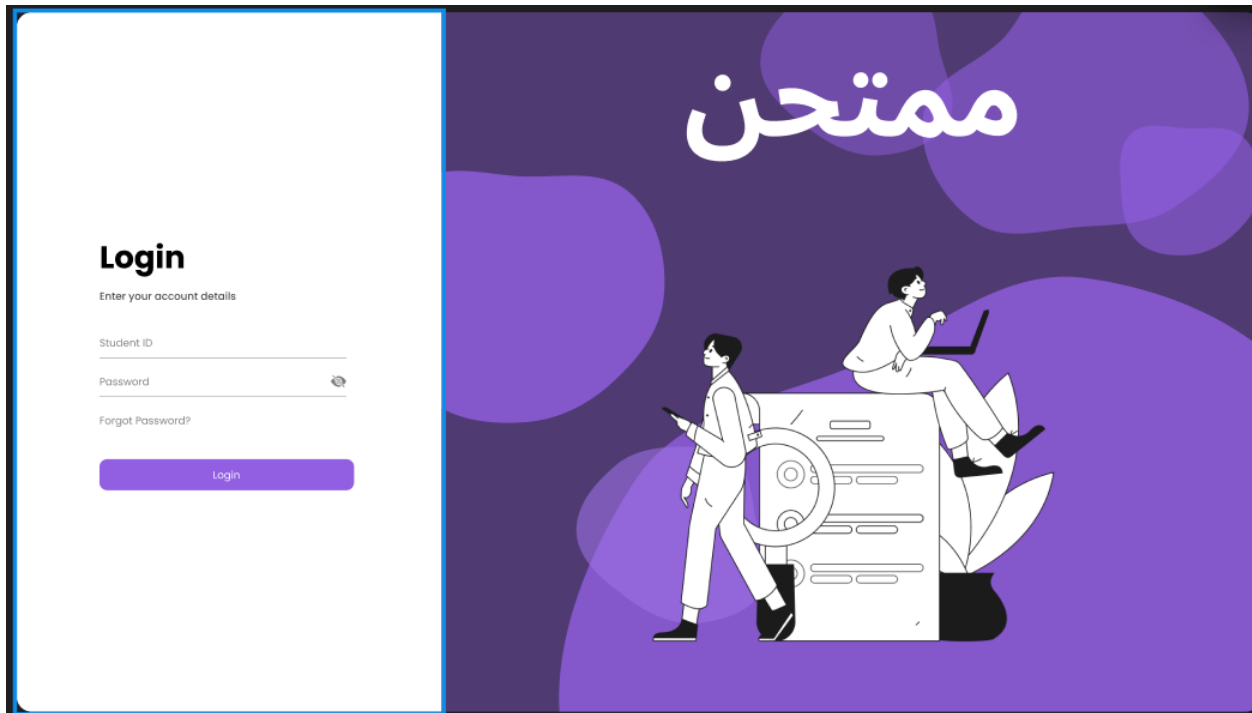
17.2 Back-End Implementation

The back-end is developed using **Flask** and **MongoDB**. **Flask** is used to create a web server that handles the application's requests, while **MongoDB** stores user data, reports, and other information in a scalable database. This back-end efficiently processes data and communicates with the front-end through secure APIs.

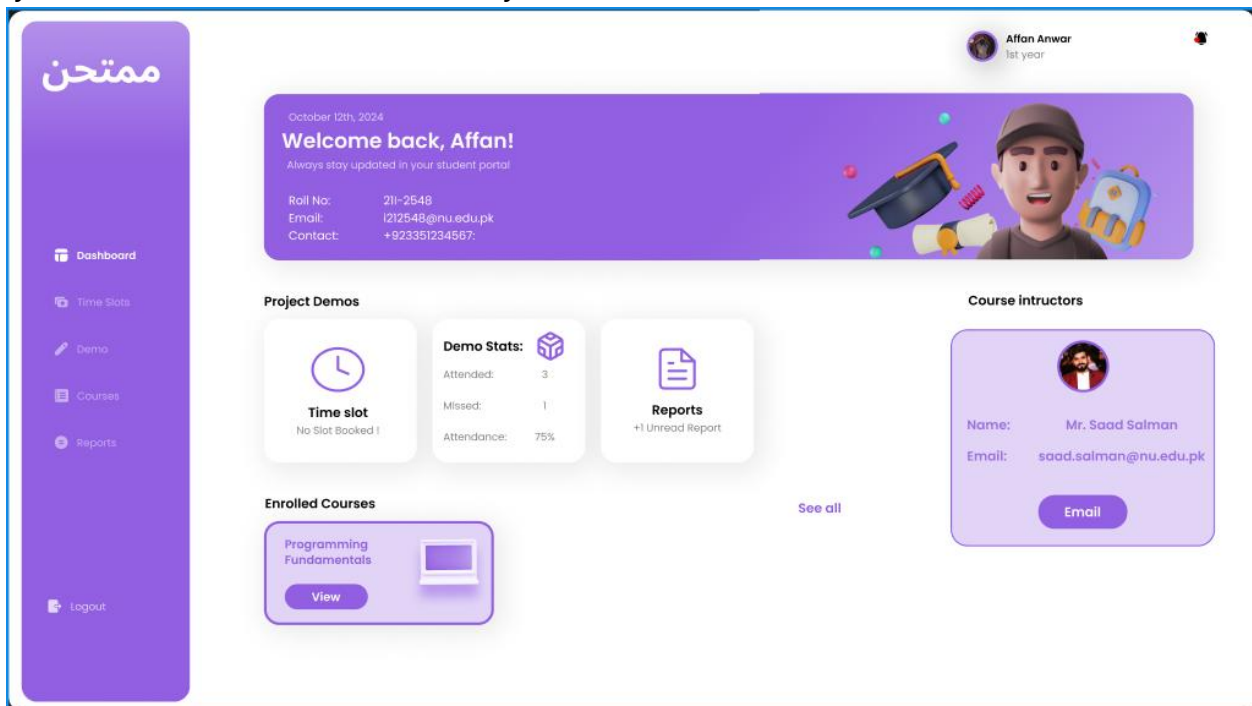
18. User Manual

1. Install the app on your desktop, when you open your application you will view a screen that tells you whether you want to login as student or teacher. Afterward you can login.

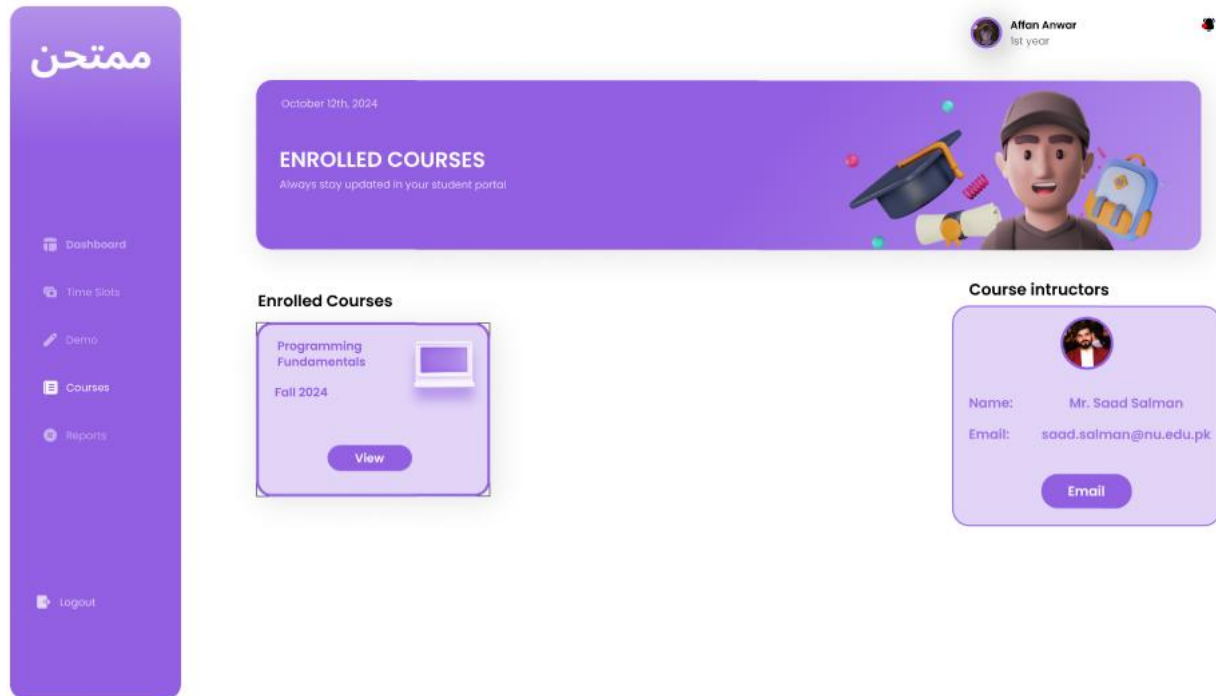




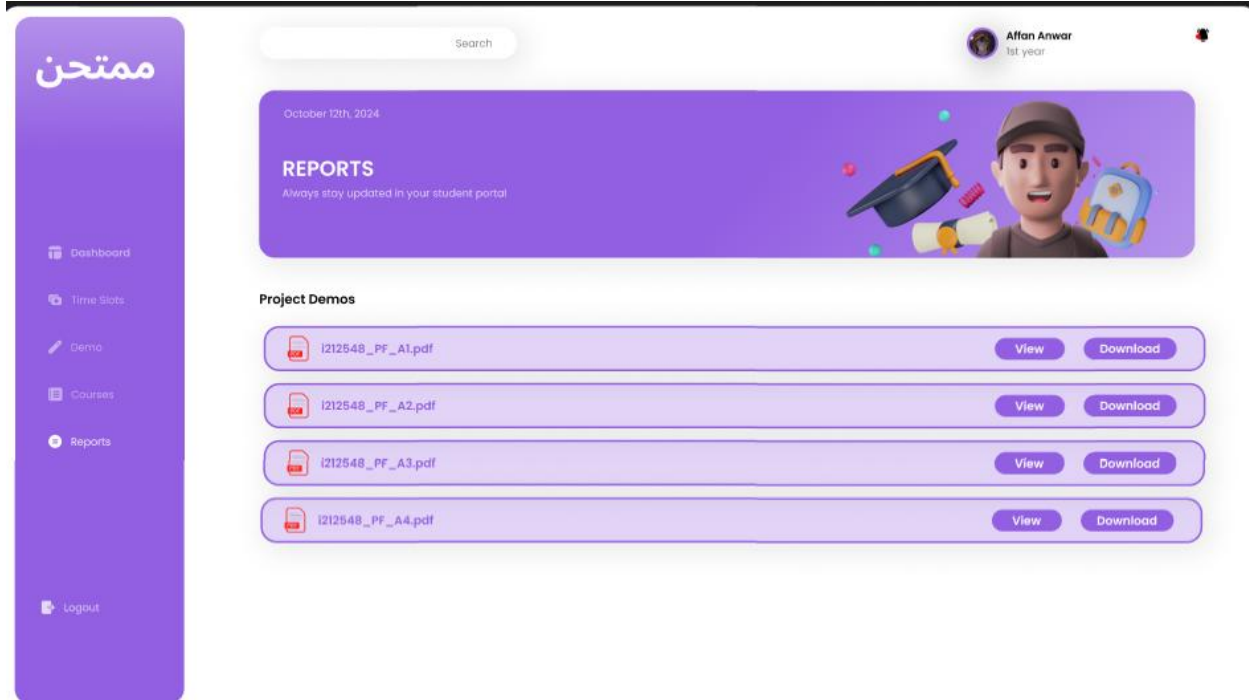
2. Now after successful login as student you will reach a dashboard where there is your profile ,your demo slot ,enrolled courses and your course instructor detail.



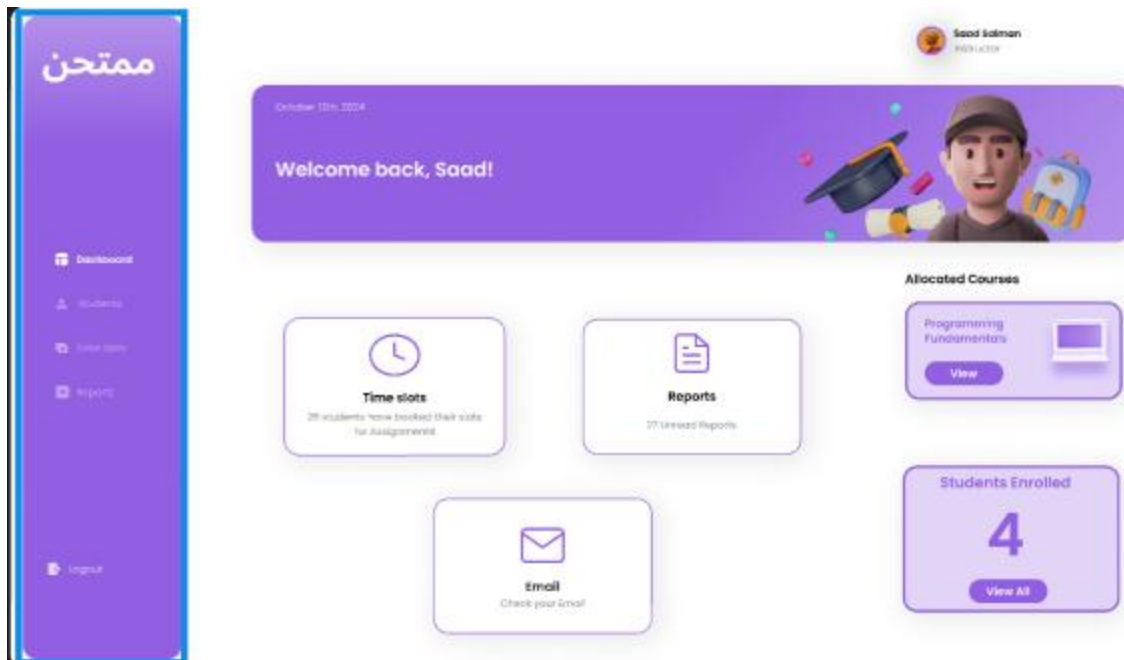
3. A student can select its own time slot according to the date available ,and this page also shows the pending demo and missed demo.



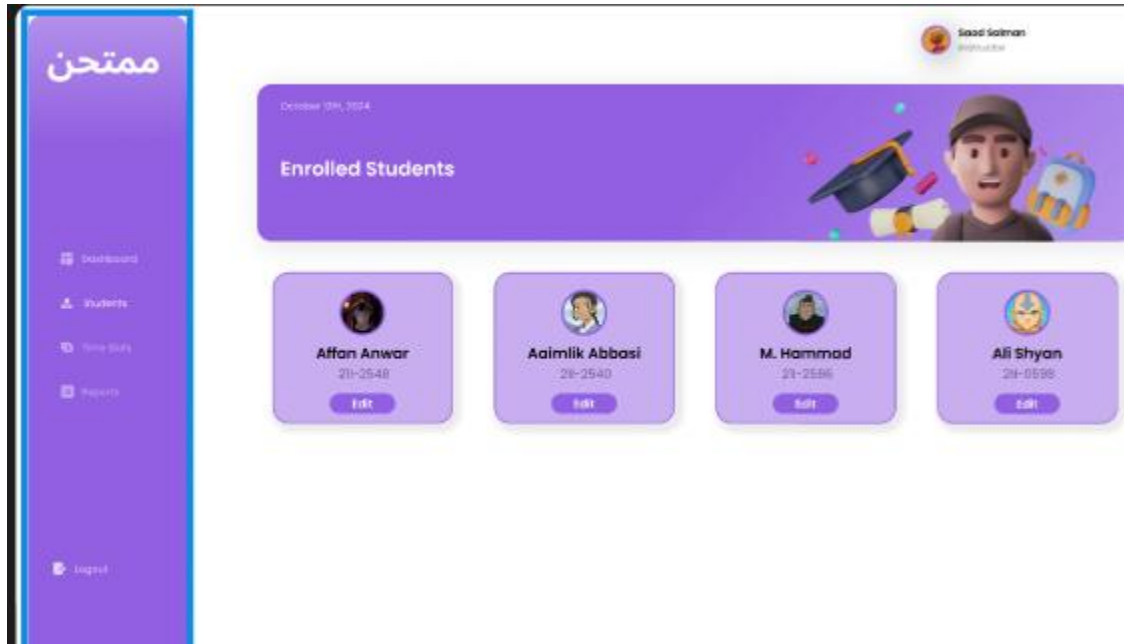
6. By clicking on Reports you can view all the reports that you have submitted on your classroom



7. If you login as teacher then this is the dashboard that will appear in front of you



8. A teacher can view all his enrolled student



9. A teacher can update timeslot of the student

18. Conclusion:

In summary, we aim to provide a standardized environment for evaluating student assignments, automating much of the demo process to ensure fairness, efficiency, and transparency. By integrating AI into the evaluation, we seek to eliminate biases, reduce manual labor, and streamline time slot scheduling, thereby reducing conflicts between students and evaluators. Our goal is to enhance the overall experience for both students and faculty, making the system more reliable and scalable for future use.